AIML-500 MACHINE LEARNING FUNDAMENTALS

Complete Course Summary – Key Takeaways

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What Is Machine Learning?

Definition: Machine Learning studies algorithms that improve automatically through experience (data), without explicit programming for every rule.

Three Core Ingredients:

- Data: Examples for learning
- Model: Mathematical structure predicting outcomes
- Learning Algorithm: Adjusts the model based on data

Key Insight: ML detects patterns in existing data and applies them to new, unseen data.

The Machine Learning Lifecycle

01

Data Preparation
60-80% of Time
- Data Collection
- Data Cleaning
- Labeling
- Feature Engineering

This is where most ML effort is spent!

02

Model Training & Tuning
10-20% of Time
- Algorithm Selection
- Hyperparameter Tuning
- Model Optimization
- Iterative Refinement

The actual "machine learning" step.

03

Evaluation & Deployment
10-20% of Time
- Performance Testing
- Validation Metrics
- Production Deployment
- Monitoring

Key Takeaway: 80% is data work!

Supervised vs. Unsupervised Learning

04

Supervised Learning
• Data: Labeled (inputs + outputs)
• Goal: Predict output from input
• Types: Regression, Classification
• Use Cases: Churn prediction, fraud detection, spam filtering
• Key Insight: Model learns from known answers to predict new outcomes

04

Unsupervised Learning
• Data: Unlabeled (input only)
• Goal: Discover hidden patterns
• Types: Clustering, Dimensionality Reduction
• Use Cases: Customer segmentation, anomaly detection, data compression
• Key Insight: Model finds structure without predefined answers

Regression vs. Classification

05

Regression

- Output: Continuous number
- Question: "How much?"
- Example: House price prediction
- Algorithms: Linear regression, decision trees
- Metrics: MSE, MAE, R^2

05

Classification

- Output: Discrete category
- Question: "Which one?"
- Example: Will house sell in 30 days? Yes/No
- Algorithms: Logistic regression, random forest
- Metrics: Accuracy, Precision, Recall, F1

Key Algorithms Overview

Linear Regression

Logistic Regression

Decision Trees

Random Forest

SVM & Neural Networks

The Bias-Variance Tradeoff

01

Bias (Underfitting)
Model is too simple and misses underlying patterns in the data.

- High error on training data
- High error on test data
- Fails to capture relationships
- Solution: Increase model complexity

02

Variance (Overfitting)
Model is too complex and memorizes noise instead of learning patterns.

- Low error on training data
- High error on test data
- Poor generalization
- Solution: Regularization, more data

03

The Sweet Spot
Balanced complexity that captures true patterns without memorizing noise.

- Good training accuracy
- Good test accuracy
- Strong generalization
- Goal: Minimize total error

Training, Validation & Test Sets

01

Training Set (60-70%)
The model learns patterns from this data. This is where the algorithm adjusts its parameters to minimize prediction errors.
Key: Largest portion of your data goes here.

02

Validation Set (15-20%)
Used to tune hyperparameters and prevent overfitting. Helps you compare different model configurations.
Key: Guides model selection without touching test data.

03

Test Set (15-20%)
Final unbiased evaluation of model performance. Simulates real-world, unseen data.
Critical Rule: Never use test set until the final step!

Cross-Validation

Cross-validation is a technique to better estimate model performance on limited data by using all available data for both training and validation.

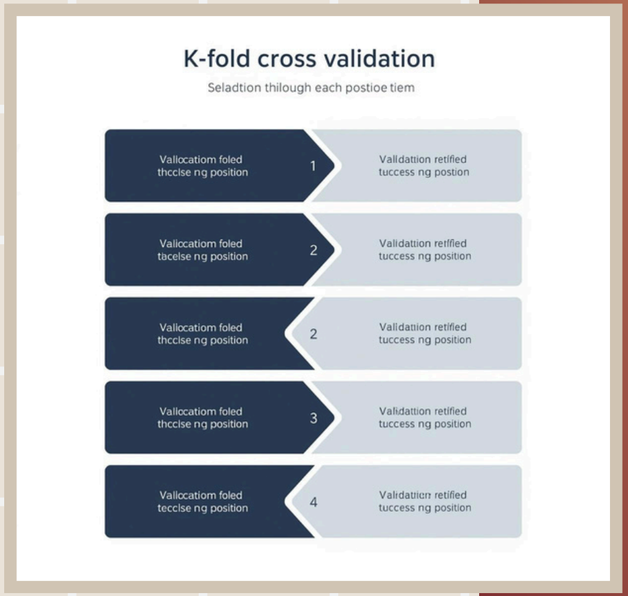
K-Fold CV Steps:

1. Split data into K equal folds
2. Train on K-1 folds, validate on remaining fold
3. Repeat K times with different validation fold
4. Average all results for final estimate

Key Benefits:

- More reliable performance estimates
- Uses all data for training and validation
- Reduces effects of lucky/unlucky data splits

Common K Values: 5 or 10 folds are standard choices, balancing computational cost with estimate reliability.



Feature Engineering – The Secret Weapon

Scaling/Normalization

Encoding Categorical Variables

Interaction Features

Polynomial Features

Domain-Specific Features

Evaluation Metrics – Regression

MSE (Mean Squared Error)

MAE (Mean Absolute Error)

R^2 (R-Squared)

RMSE (Root Mean Squared Error)

Classification Metrics

Accuracy: Overall correctness (TP+TN)/Total

Precision: Correct positives / predicted positives

Recall: Correct positives / actual positives

F1 Score: Harmonic mean of Precision & Recall

Choose metric based on error costs

Overfitting vs. Underfitting – Visual Guide

01

Underfitting
Model too simple, misses important patterns in the data.
• Poor training accuracy
• Poor test accuracy
• High bias, low variance
• Solution: Add features, increase model complexity

02

Just Right
Model captures true underlying patterns without memorizing noise.
• Good training accuracy
• Good test accuracy
• Balanced bias-variance
• Goal: Optimal generalization to new data

03

Overfitting
Model too complex, memorizes training data including noise.
• Excellent training accuracy
• Poor test accuracy
• Low bias, high variance
• Solution: Regularization, more data, simpler model

Regularization – Preventing Overfitting

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L1 Regularization (Lasso)
• Penalizes absolute value of coefficients
• Can zero out features completely
• Performs automatic feature selection
• Best when few features are truly important
Elastic Net
• Combines L1 and L2 penalties
• Balances feature selection with stability

L2

L2 Regularization (Ridge)
• Penalizes squared coefficients
• Shrinks coefficients but keeps all features
• More stable with correlated features
• Best when most features contribute
Other Techniques
• Dropout (neural networks)
• Early stopping
• Data augmentation

Clustering – Unsupervised Learning

01

K-Means Clustering
Most popular clustering algorithm. Fast and scalable for large datasets. Requires specifying K (number of clusters) upfront. Works best with spherical, evenly-sized clusters. Uses iterative centroid optimization.

02

Hierarchical Clustering
Builds a tree-like dendrogram structure. No need to specify K in advance. Can cut tree at any level for desired clusters. Two approaches: agglomerative (bottom-up) or divisive (top-down). Good for exploring data structure.

03

DBSCAN
Density-based spatial clustering. Detects clusters of arbitrary shapes. Automatically identifies outliers as noise points. No need to specify number of clusters. Requires density parameters (eps, min_samples).
Applications: Customer segmentation, anomaly detection, image compression

Dimensionality Reduction – PCA

Problem: Too many features cause overfitting, slow training, and difficult visualization.

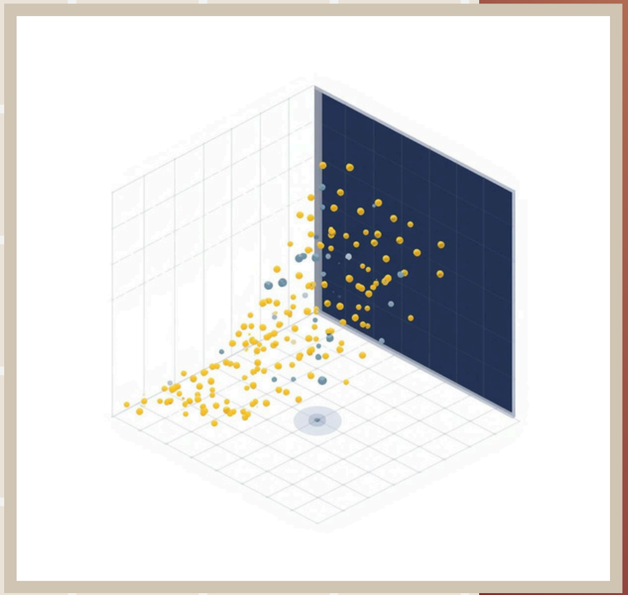
PCA Steps:

1. Standardize data
2. Compute covariance matrix
3. Find principal components (eigenvectors)
4. Project data onto top K components

Benefits:

- Reduces dimensionality and noise
- Helps visualization (2D/3D)
- Improves model training speed

Trade-off: Some information and interpretability lost.



keaways for Exams & Interviews

Supervised vs. Unsupervised Learning

Bias-Variance Tradeoff

Train/Validation/Test Split

Feature Engineering Importance

The 80% Rule – Data Prep Dominance